Short note and example on absorbing and supplying of power for elements in a circuit

Basic:

If Current (Positive or negative) flows in the positive side of an element, the power formula is

$$P = VI$$

If Current (Positive or negative) flows in the negative side of an element, the power formula is

$$P = -VI$$

After Calculation, if the value of power is positive, then the element is absorbing power. And if the value of power is negative, then the element is supplying power.

Let us clarify the calculation of power of the -6V voltage source (Marked inside the blue box)

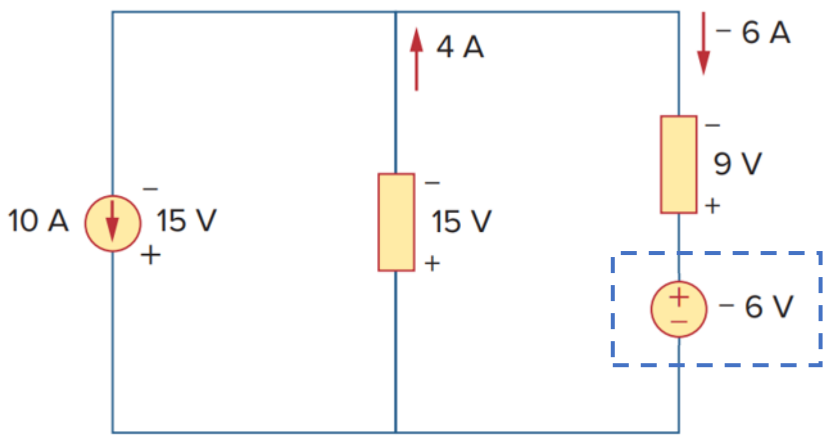


Figure 1

You can calculate the power of -6V voltage source in **3 methods**

1st method:

Keeping the circuit as it is. So, the -6A current is flowing in the (Positive) side of the voltage source. So, the power equation becomes,

$$P = VI$$

Putting $I = -6A$ and $V = -6V$ in the equation,

$$P = (-6A) \times (-6V)$$

$$P = 36\,W$$

As the value is positive, voltage source is absorbing the power

For the 2nd and 3rd methods, let us represent the current direction in a conventional way (without the minus sign). In the Figure 1, On the right side, the current direction is given downwards for -6A. Actually, 6A current is flowing from bottom to top, like in the Figure 2 below.

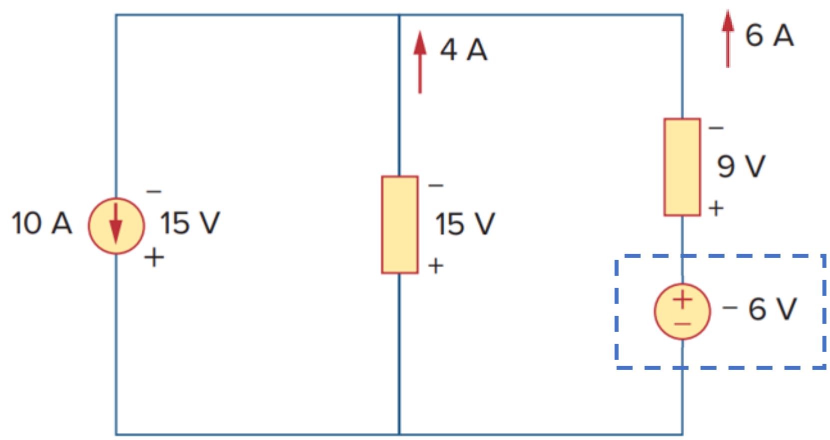


Figure 2

2nd method:

As per Figure 2, the 6A current is flowing in the (Negative) side of the voltage source. So, the power equation becomes,

$$P = -VI$$

Putting $I = 6A$ and $V = -6V$ in the equation,

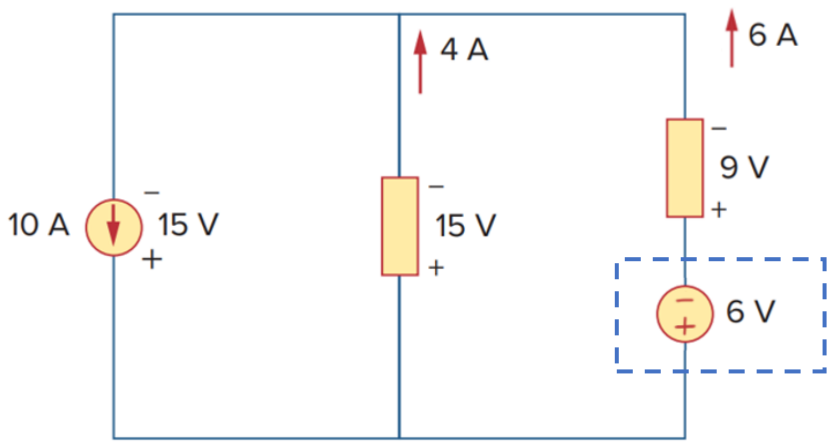
$$P = -(6A \times (-6V))$$

$$P = 36\,W$$

As the value is positive, voltage source is absorbing the power

3rd method:

Reverse the voltage source so that it becomes 6V (For calculation purpose only. You do not need to show it; just draw a rough circuit for yourself). The circuit will look like Figure 3.



So, the 6A current is flowing in the (Positive) side of the voltage source. So, the power equation becomes,

$$P = VI$$

Putting $I = 6A$ and $V = 6V$ in the equation,

$$P = 6A \times 6V$$

$$P = 36\,W$$

As the value is positive, voltage source is absorbing the power